

Artificial Intelligence

Course Information

- **Course Code:** CSC462 Course
- **Title:** Artificial Intelligence
 - **Credit Hours:** 4 (3,1)

Text and Reference Books

- **Textbook:** 1. Artificial Intelligence: A Modern Approach, Russell, S., and Norvig, P., Pearson, 2020.
- **Reference Book:** 1. Artificial Intelligence Basics: A Non-Technical Introduction, Taulli, T., Apress, 2019.



Artificial Intelligence Search

Agents Informed search

Topics Covered: Heuristic Function, Greedy Best First Search, and A* search.

Reading Material: AIMA: Ch3

Informed search

- In **Artificial Intelligence**, an **informed search algorithm** uses **heuristic information** to guide the search towards the goal **more efficiently** than an **uninformed (blind) search**.
This extra knowledge helps reduce **search time**, **memory usage**, and **irrelevant exploration**.

Heuristic Function

- A **heuristic function**, often written as **$h(n)$** , is a **guide** that helps an AI algorithm **estimate how close a state is to the goal**. It provides ***extra knowledge*** about the problem to **improve search efficiency**.
- The value of the heuristic function is always **positive**.

Think of it like **Google Maps**:

- $g(n)$ = Distance you **have already traveled**.
- $h(n)$ = **Estimated distance left** to reach your destination.
- $f(n)$ = Total estimated cost $\rightarrow$ **$f(n) = g(n) + h(n)$** .

Heuristic Function

Purpose of Heuristic Function

- **Guides the search** towards the most promising paths.
- **Reduces time and memory usage** by avoiding unnecessary exploration.
- Helps **informed search algorithms** like A*, Greedy Best-First, etc., work efficiently.
- Can **determine optimality** : if the heuristic is accurate, the algorithm can find the best solution.

*A * Search Algorithm*

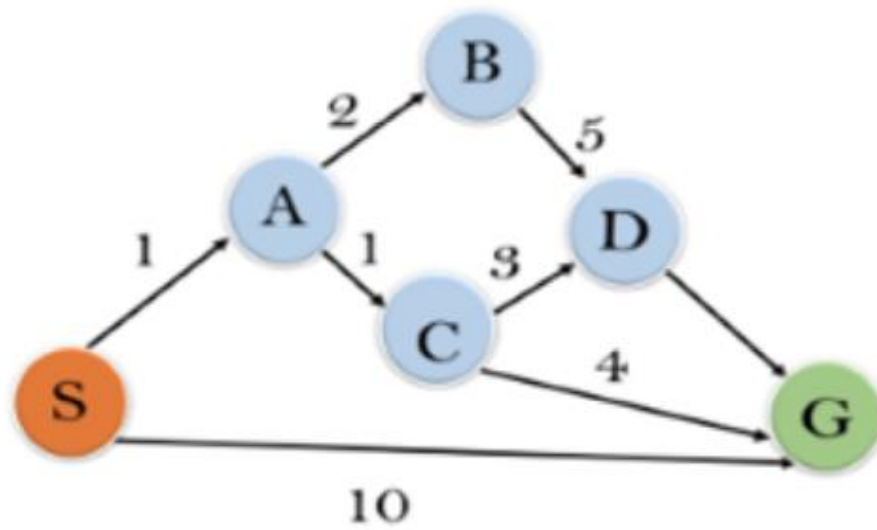
- The A * algorithm is a popular pathfinding and graph traversal algorithm used to find the **shortest path between two nodes**.
- uses a **cost function** to evaluate nodes and prioritizes the most promising paths.
- **Cost Function:**
- A* uses a cost function $f(n)$ to evaluate nodes:
- **$f(n)=g(n)+h(n)$**
- **$g(n)$:** The actual cost to reach the current node n from the start node.
- **$h(n)$:** The heuristic estimate of the cost to reach the goal node from node n .
- **$f(n)$:** The total estimated cost of the path through node n .

Where A* Can Be Used



Search Type	Used?
Tree Search	Yes
Graph Search	Yes (with a closed list)
Directed Graphs	Yes
Weighted Graphs	Yes
Negative Weights	No

Example 1:



State	$h(n)$
S	5
A	3
B	4
C	2
D	6
G	0

$$f(n) = g(n) + h(n)$$

Step 1: Initialization

$$S \rightarrow h(5) = 5$$

iteration 1: $S \rightarrow A =$

$$f(n) = g(n) + h(n)$$

$$f(n) = 1 + 3 = \boxed{4} \checkmark$$

$S \rightarrow G$

$$f(n) = g(n) + h(n)$$

$$f(n) = 10 + 0 = \boxed{10}$$

iteration 2: $S \rightarrow A \rightarrow B$

$$f(n) = g(n) + h(n)$$

$$f(n) = (1+2) + 4$$

$$\boxed{f(n) = 7}$$

$S \rightarrow A \rightarrow C$

$$f(n) = g(n) + h(n)$$

$$f(n) = (1+1) + 2$$

$$f(n) = \boxed{4} \checkmark$$

$S \rightarrow G$

$$10 + 0 = \boxed{10}$$

iteration 3:

$S \rightarrow A \rightarrow C \rightarrow D$

$$f(n) = g(n) + h(n)$$

$$f(n) = (1+1+3) + 6 = \boxed{11}$$

$S \rightarrow A \rightarrow B = f(n) = g(n) + h(n)$

$$f(n) = (1+2) + (4) = \boxed{7}$$

$S \rightarrow G = \boxed{10}$

Page No. _____

$$S \rightarrow A \rightarrow C + c_7$$

$$f(n) = g(n) + h(n)$$

$$f(n) = (1 + 1 + 4) + 0 = 6$$

$$[f(n) = 6]$$

iteration 4:

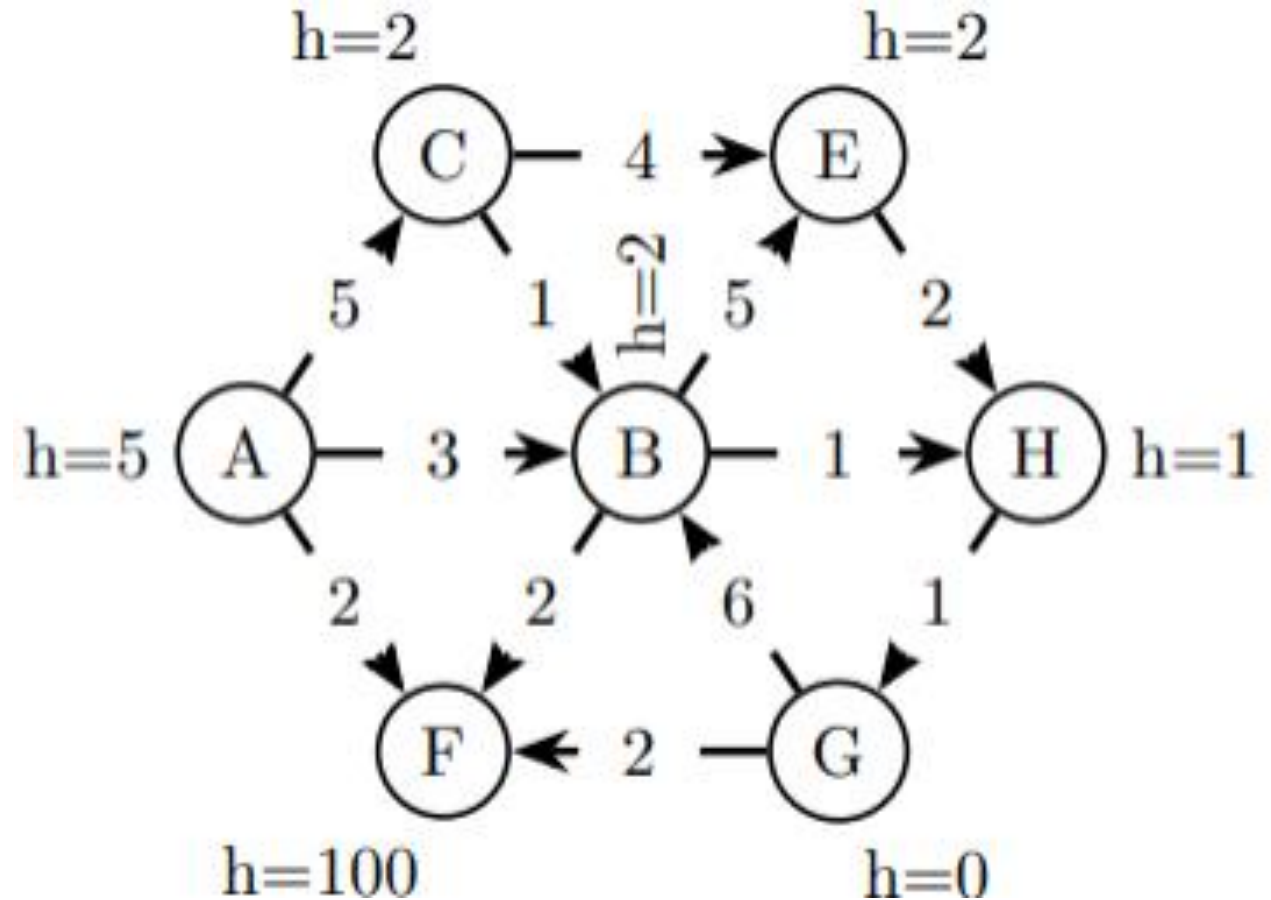
we will give the final result as

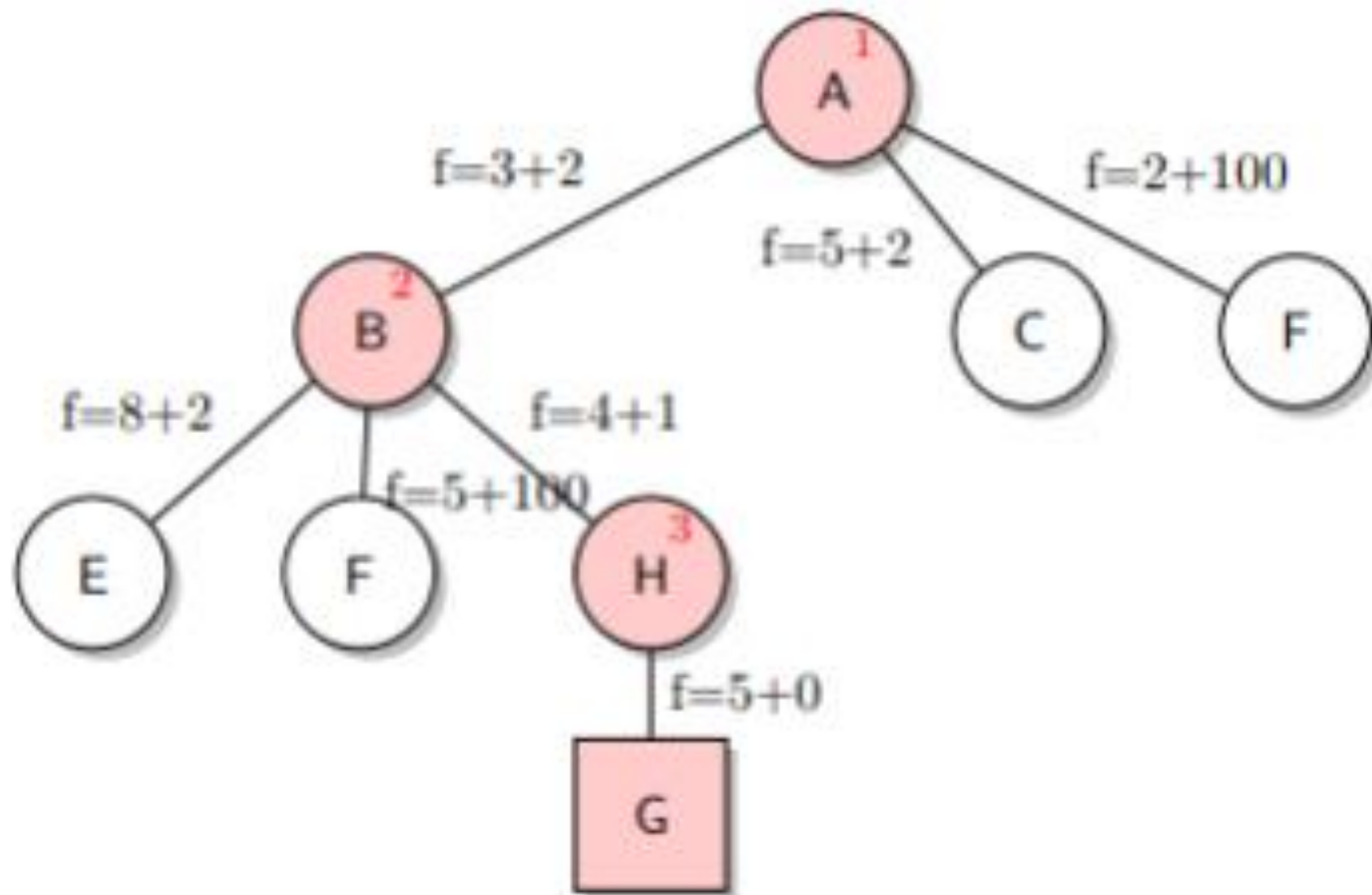
$$S \rightarrow A \rightarrow C \rightarrow c_7$$

it provide the optimal path with
cost $[6]$

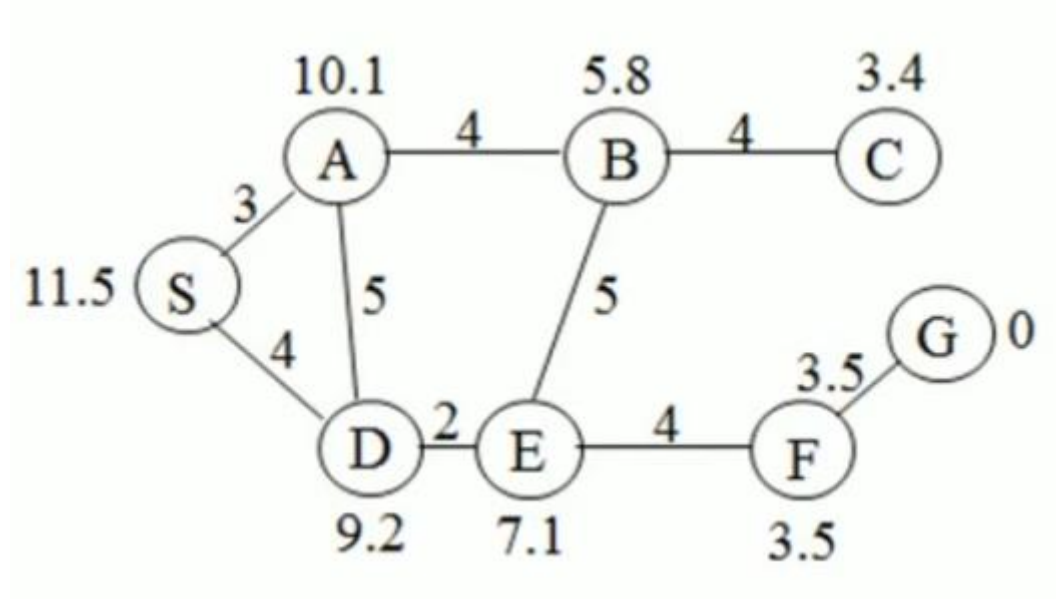
Example 2

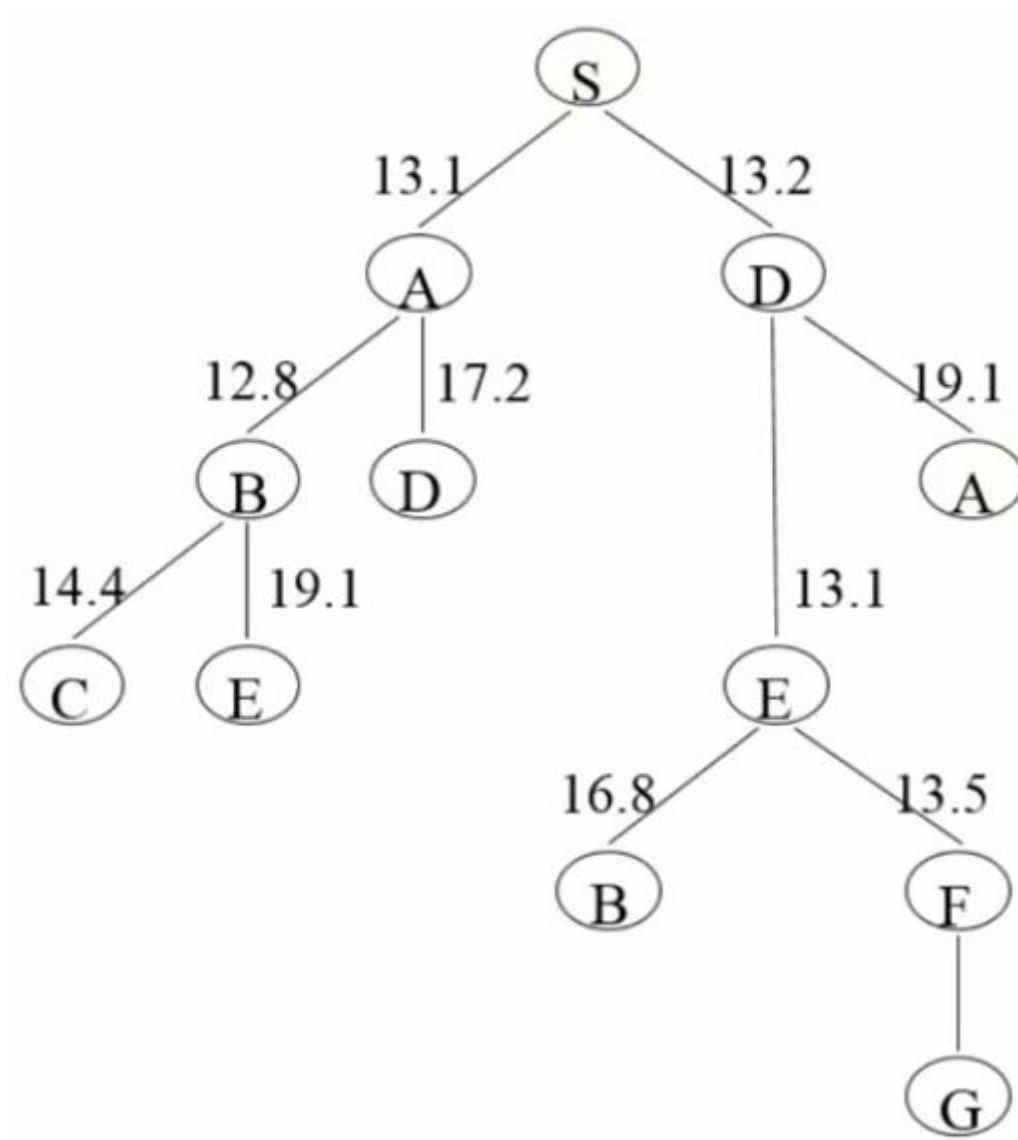
- Given the search problem defined by the following graph, **A** is the initial state and **G** is the goal state and the cost of each action is indicated on the corresponding edge.





Example 3:





Example 4:

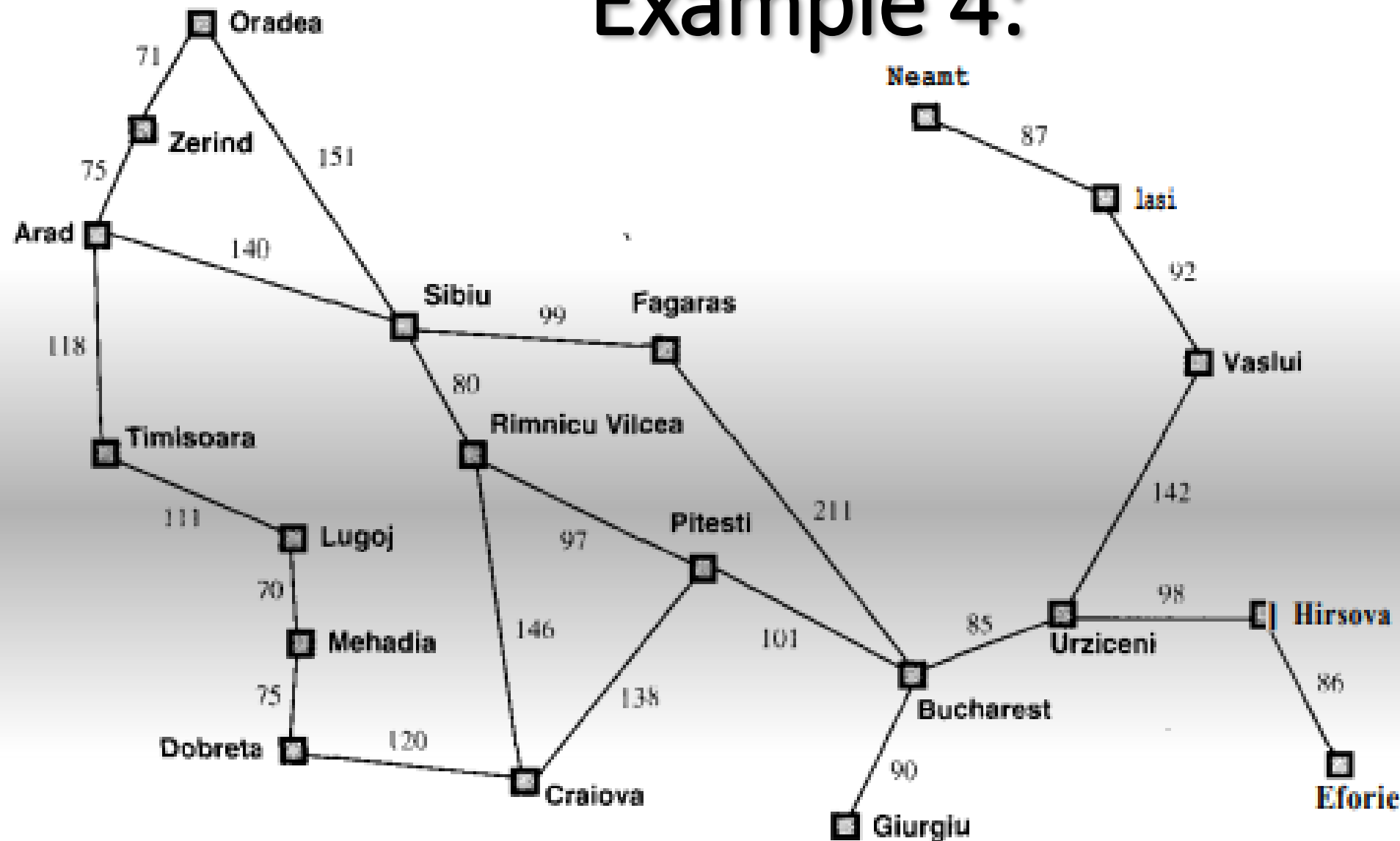


Figure 4.2 Map of Romania with road distances in km, and straight-line distances to Bucharest.

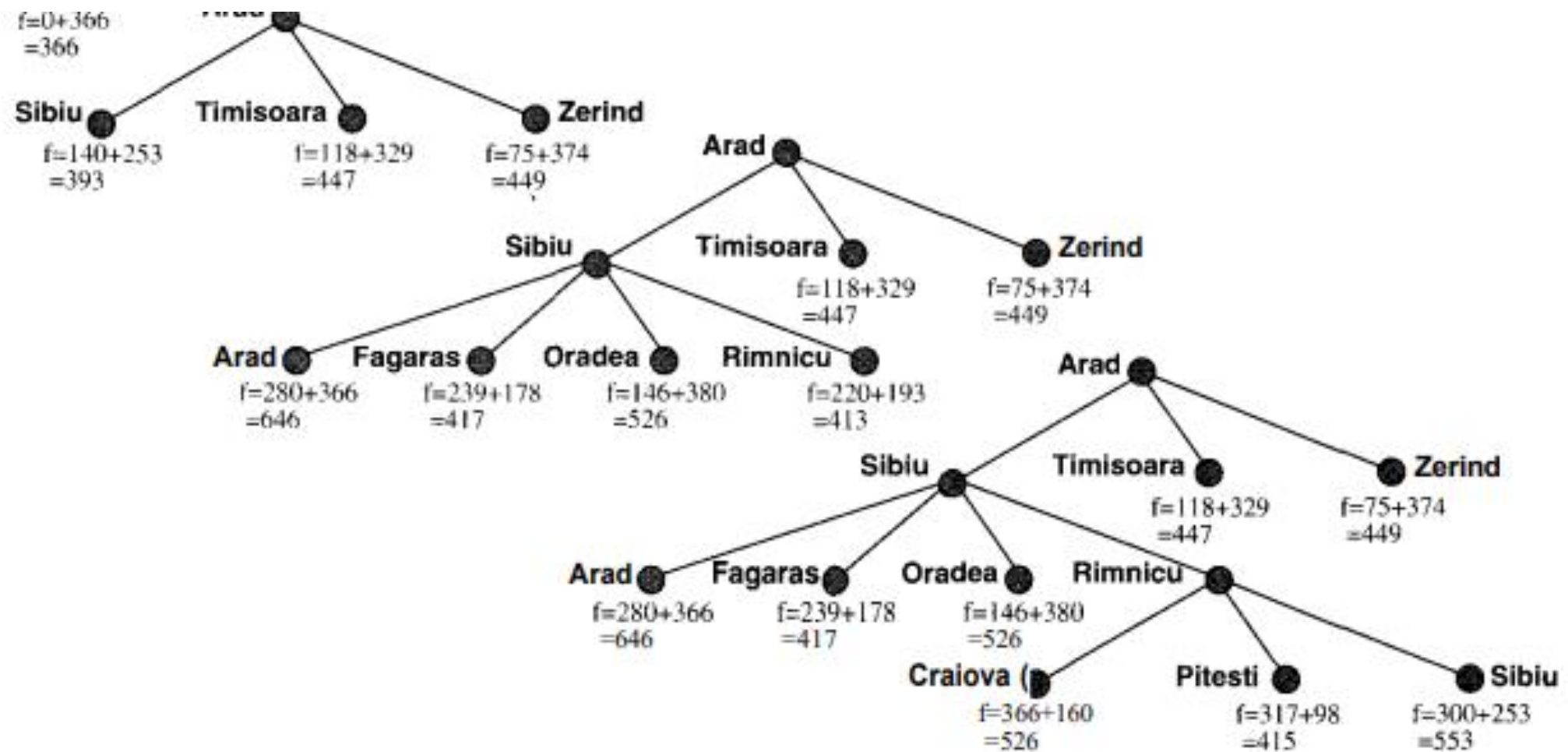


Figure 4.4 Stages in an A* search for Bucharest. Nodes are labelled with $f = g + h$. The h values are the straight-line distances to Bucharest taken from Figure 4.1.

Water Jug Problem in Artificial Intelligence



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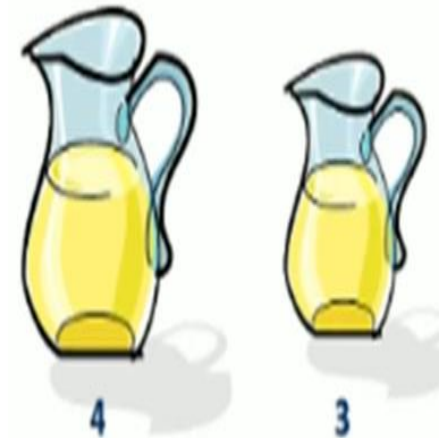


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Water Jug Problem in Artificial Intelligence

Water Jug Problem Definition,

- You are given two jugs, a 4-liter one and a 3-liter one. Neither has any measuring markers on it. There is a pump that can be used to fill the jugs with water. How can you get exactly 2 liters of water into a 4-liter jug.



Representation of water Jug Problem in terms of state-space search,

- State: (x, y)
- where x represents the quantity of water in a 4-liter jug and y represents the quantity of water in a 3-liter jug.
- That is, $x = 0, 1, 2, 3, \text{ or } 4$ $y = 0, 1, 2, 3$
- Start state: $(0, 0)$
- Goal state: $(2, n)$ for any n .
- Here we need to start from the start state and end up in a goal state.

1	$(x, y) \text{ is } X < 4 \rightarrow (4, Y)$	Fill the 4-liter jug
2	$(x, y) \text{ if } Y < 3 \rightarrow (x, 3)$	Fill the 3-liter jug
3	$(x, y) \text{ if } x > 0 \rightarrow (x-d, y)$	Pour some water out of the 4-liter jug.
4	$(x, y) \text{ if } Y > 0 \rightarrow (x, y-d)$	Pour some water out of the 3-liter jug.
5	$(x, y) \text{ if } x > 0 \rightarrow (0, y)$	Empty the 4-liter jug on the ground
6	$(x, y) \text{ if } y > 0 \rightarrow (x, 0)$	Empty the 3-liter jug on the ground
7	$(x, y) \text{ if } X+Y \geq 4 \text{ and } y > 0 \rightarrow (4, y-(4-x))$	Pour water from the 3-liter jug into the 4-liter jug until the 4-liter jug is full
8	$(x, y) \text{ if } X+Y \geq 3 \text{ and } x > 0 \rightarrow (x-(3-y), 3)$	Pour water from the 4-liter jug into the 3-liter jug until the 3-liter jug is full.
9	$(x, y) \text{ if } X+Y \leq 4 \text{ and } y > 0 \rightarrow (x+y, 0)$	Pour all the water from the 3-liter jug into the 4-liter jug.
10	$(x, y) \text{ if } X+Y \leq 3 \text{ and } x > 0 \rightarrow (0, x+y)$	Pour all the water from the 4-liter jug into the 3-liter jug.
11	$(0, 2) \rightarrow (2, 0)$	Pour the 2-liter water from the 3-liter jug into the 4-liter jug.
12	$(2, Y) \rightarrow (0, y)$	Empty the 2-liter in the 4-liter jug on the ground.

The solution to Water Jug Problem in Artificial Intelligence

1. Start state = $(0, 0)$
2. Loop until the goal state $(2, 0)$ reached
 - Apply a rule whose left side matches the current state
 - Set the new current state to be the resulting state

1. Solution to Water Jug Problem in Artificial Intelligence

- $(0, 0)$ – Start State
- $(0, 3)$ – Rule 2, Fill the 3-liter jug
- $(3, 0)$ – Rule 9, Pour all the water from the 3-liter jug into the 4-liter jug.
- $(3, 3)$ – Rule 2, Fill the 3-liter jug
- $(4, 2)$ – Rule 7, Pour water from the 3-liter jug into the 4-liter jug until the 4-liter jug is full.
- $(0, 2)$ – Rule 5, Empty the 4-liter jug on the ground
- $(2, 0)$ – Rule 9, Pour all the water from the 3-liter jug into the 4-liter jug.
- Goal State reached

2. Solution to Water Jug Problem in Artificial Intelligence

- $(0, 0)$ – Start State
- $(4, 0)$ – Rule 1, Fill the 4-liter jug
- $(1, 3)$ – Rule 8, Pour water from the 4-liter jug into the 3-liter jug until the 3-liter jug is full.
- $(1, 0)$ – Rule 6, Empty the 3-liter jug on the ground
- $(0, 1)$ – Rule 10, Pour all the water from the 4-liter jug into the 3-liter jug.
- $(4, 1)$ – Rule 1, Fill the 4-liter jug
- $(2, 3)$ – Rule 8, Pour water from the 4-liter jug into the 3-liter jug until the 3-liter jug is full.